

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

STRUCTURE OF ATOM

LECTURE-3

CHEMISTRY

VISHAL SIR



Baaz
Batch
Zindabad



JKBose Wallah

Topics To Covered

① *Thompson model of atom*

- 1 RUTHERFORD MODEL OF ATOM**
- 2 ATOMIC NUMBER AND MASS NUMBER**
- 3 ISOTOPES AND ISOBAR AND ISOTONES**

REVISION

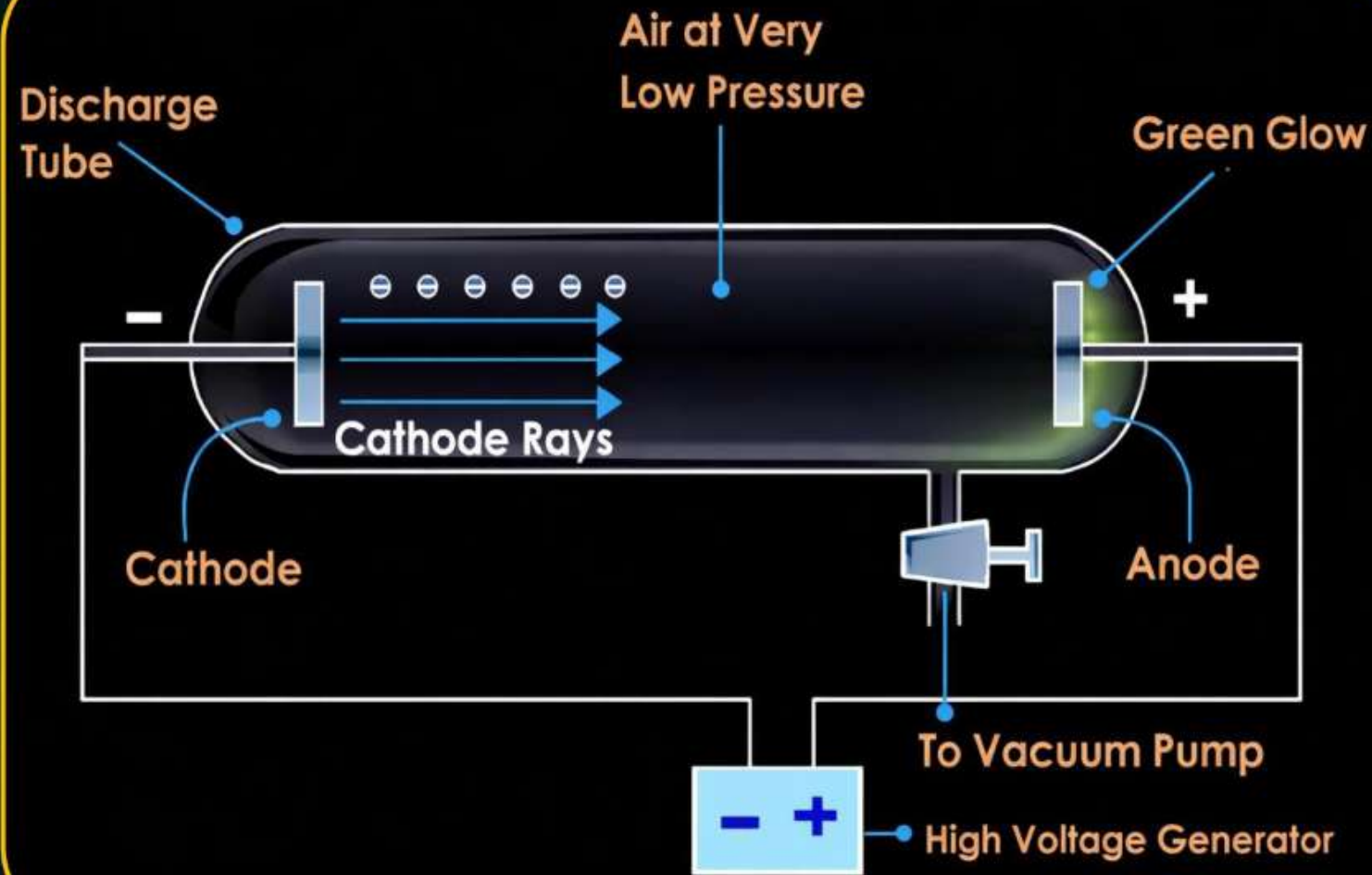


COMMUNITY

- Recorded Lecture (Kashmir event).
(Kabhi)
- Exercise Question ✓
- Homework ✓
- SHIKAYAT ✓ (dast pe bolunga).

Discovery of e^-

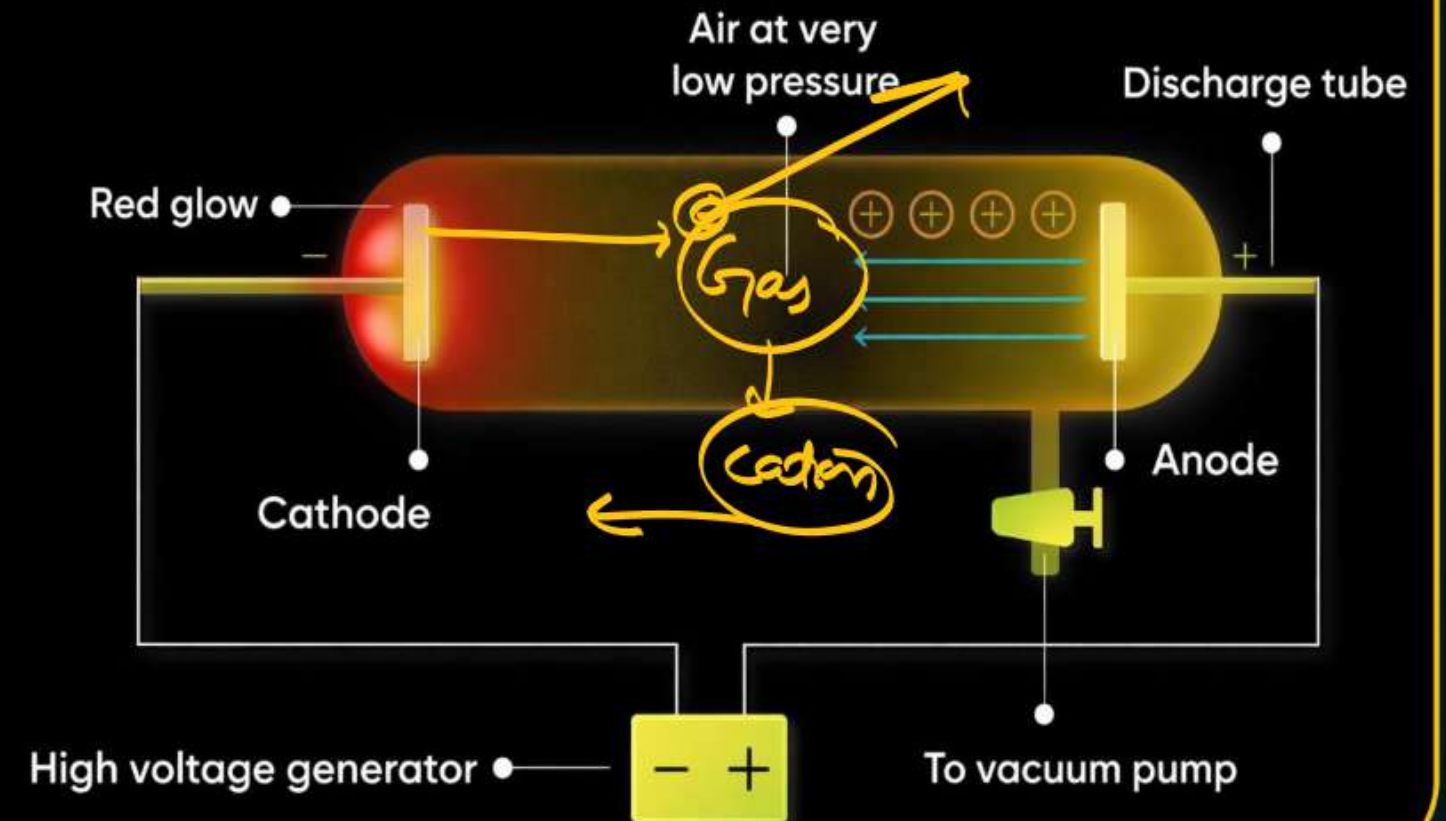
- Discharge tube experiment



Discovery of proton.

- Modified discharge tube experiment.

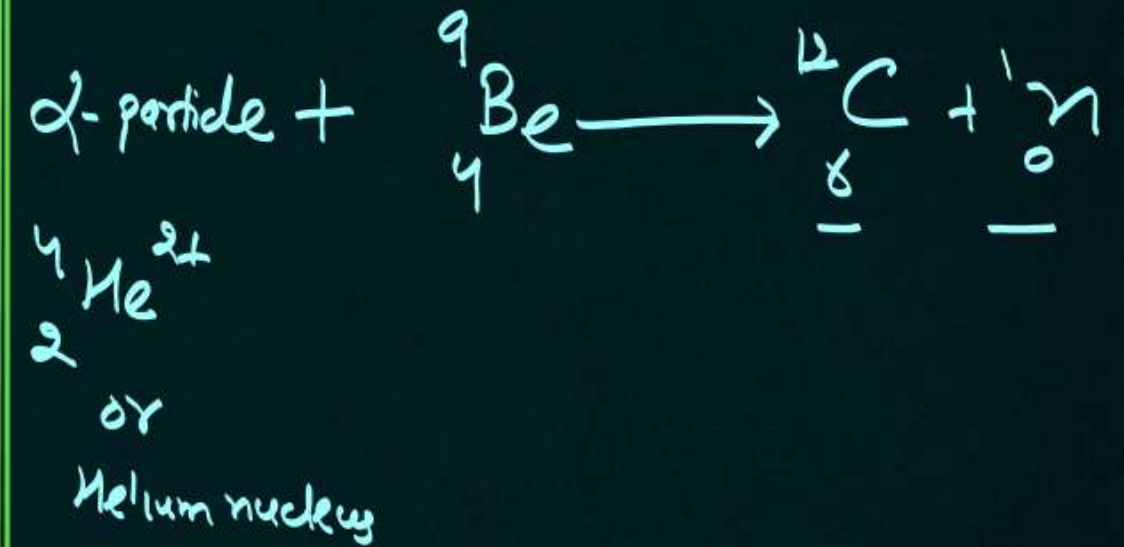
PRODUCTION OF ANODE RAYS



	Relative charge	Charge	mass	$\frac{e}{m}$
e	-1	$1.6 \times 10^{-19} \text{ C}$	$9.1 \times 10^{-31} \text{ kg}$	$1.75 \times 10^{11} \text{ C/kg}$ Neutron.

p	+1	$1.6 \times 10^{-19} \text{ C}$	$1.67 \times 10^{-27} \text{ kg}$	$9.5 \times 10^7 \text{ C/kg}$
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n	0	0	11	0
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QUESTION



Home work sha.

Calculate the number of electrons which will together weigh one gram.

Sol:-

$$9.1 \times 10^{-28} \text{ g has } = 1 \bar{e}$$

$$1 \text{ g has } = \frac{1}{9.1 \times 10^{-28}} \bar{e}$$

$$= \frac{0.1098}{10^{-28}} \bar{e}$$

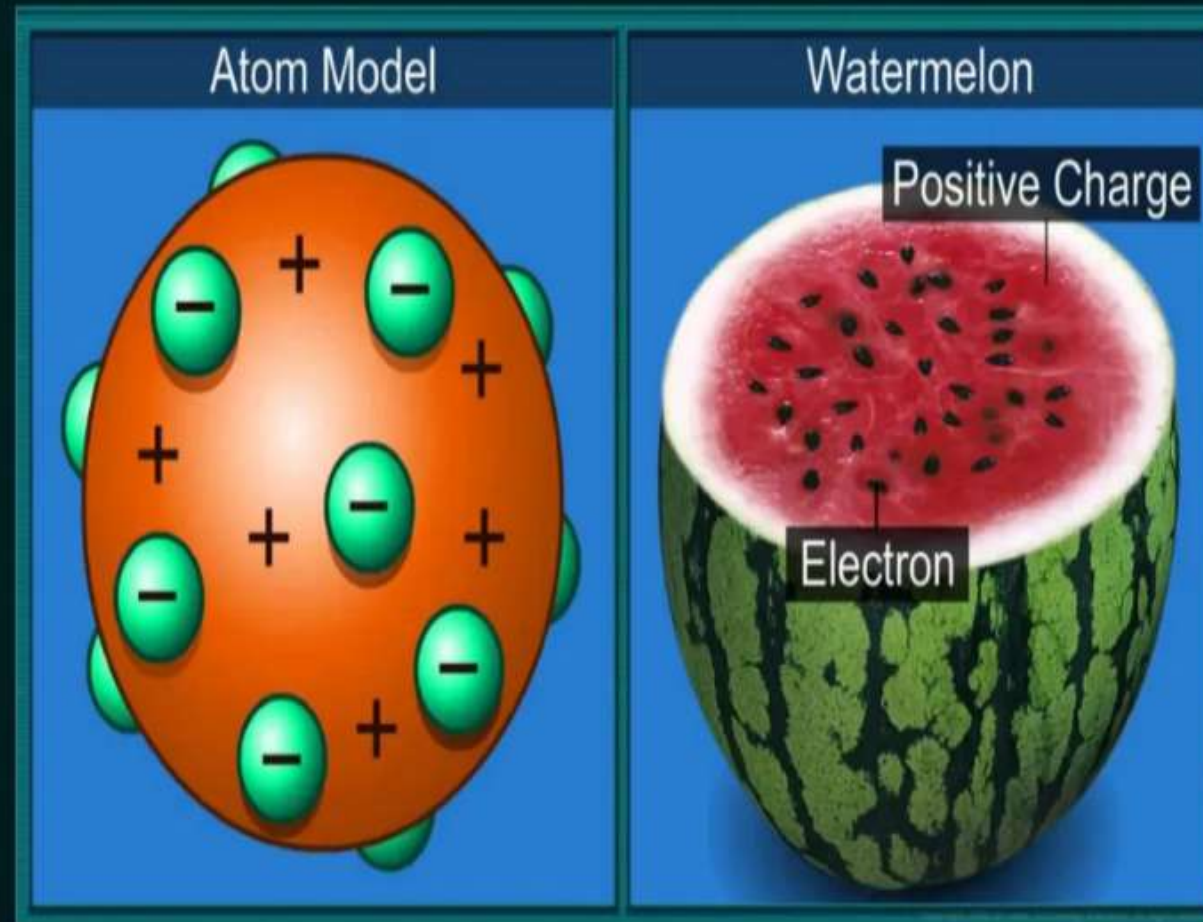
$$= 0.1098 \times 10^{28} \bar{e}$$

$$= 0.1098 \times 10^1 \times 10^{27}$$

$$= 1.098 \times 10^{27} \bar{e}$$

$$\begin{aligned} 1 \bar{e} &= 9.1 \times 10^{-31} \text{ Kg} \\ &= 9.1 \times 10^{-31} \times 1000 \text{ g} \\ &= 9.1 \times 10^{-31} \times 10^3 \\ &= 9.1 \times 10^{-28} \text{ g} \end{aligned}$$

THOMSON'S ATOMIC MODEL



According to J. J. Thompson, atom is a sphere of positive charge and negatively charged e^- are embedded in it.

- Atom is stable because the magnitude of repulsive forces is equal to the magnitude of attractive forces.
 - This model of atom is also known as Watermelon model, seeds are compared with electron & other part was considered as sphere of +ve charge.
 - Other names are plum-pudding model.
- LIMITATION → This model was discarded, because it was unable to explain Rutherford scattering experiment.

Mass	Relative Mass	Charge	Relative charge	Specific charge
0.000546 amu 9.1094×10^{-31} kg	$\frac{1}{1837}$	-602×10^{-19} C	-1	$\frac{1.76 \times 10^{11}}{\text{C/kg}}$
1.00728 amu 1.673×10^{-27} kg	1	1.602×10^{-19} C	1	$\frac{9.58 \times 10^7}{\text{C/kg}}$
1.008665 amu 1.675×10^{-27} kg	1	0	0	0

RUTHERFORD MODEL OF ATOM

(DISCOVERY OF NUCLEUS)

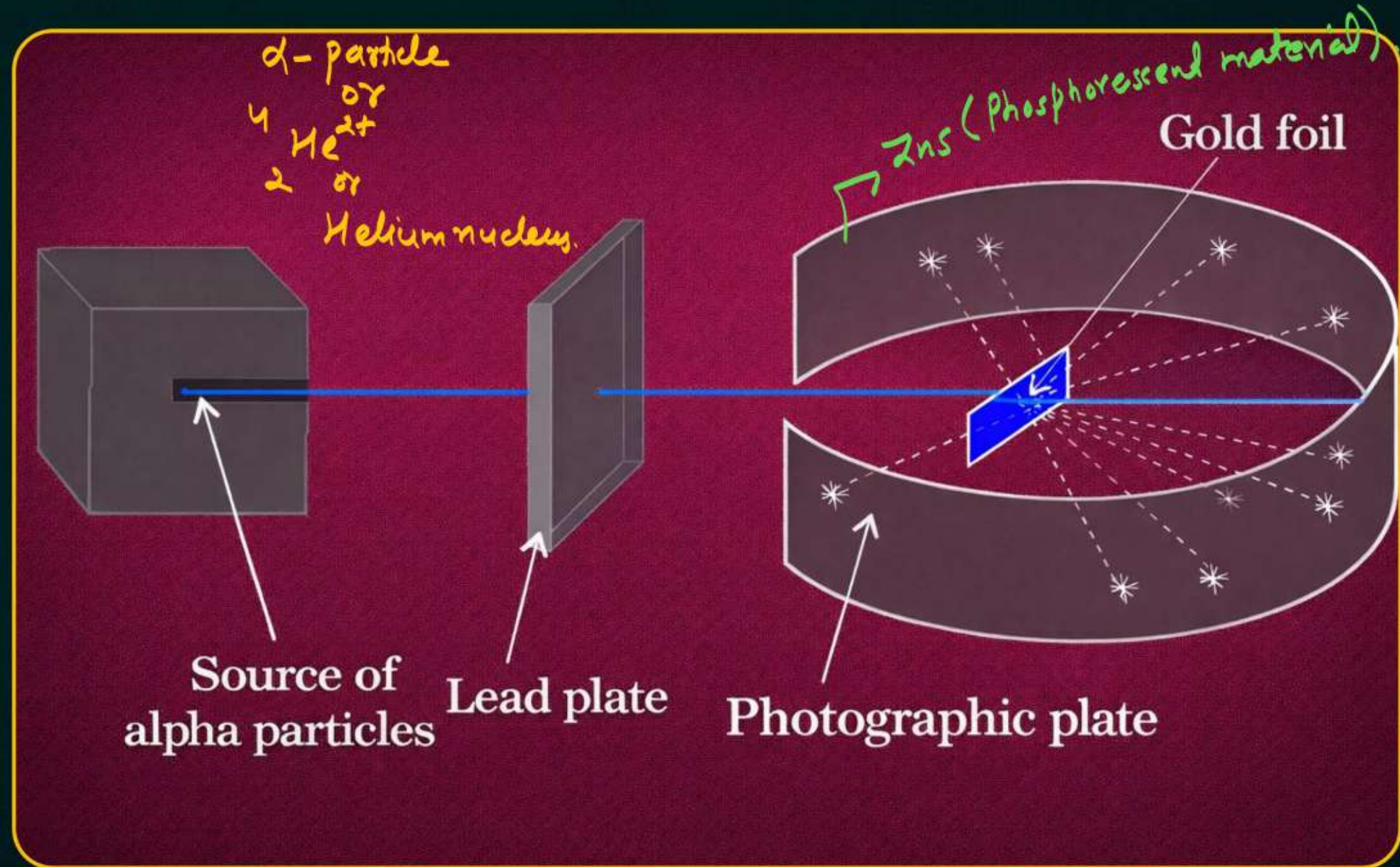


A) Rutherford Scattering Experiment :->

Rutherford bombarded a stream of α -particle on a thin sheet of Gold.

Gold foil was surrounded by a phosphorescent material (ZnS).

When any particle strikes the ZnS, it glows

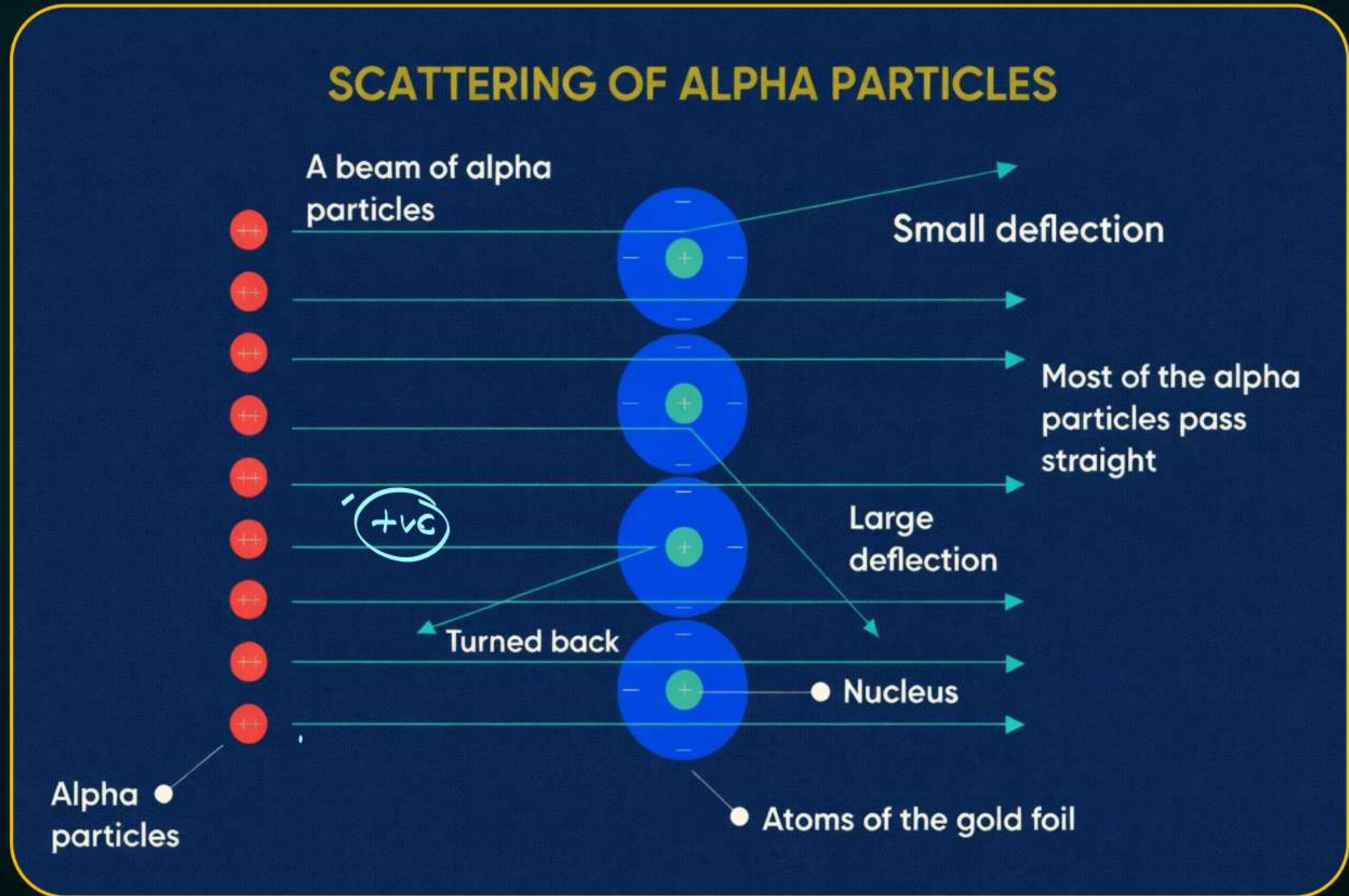


RUTHERFORD MODEL OF ATOM



OBSERVATIONS:

- Most of the α -particles were passed through the Gold foil.
- Few α -particles were deviated.
- Very few α -particles (one in 20,000), were sent back 180° .

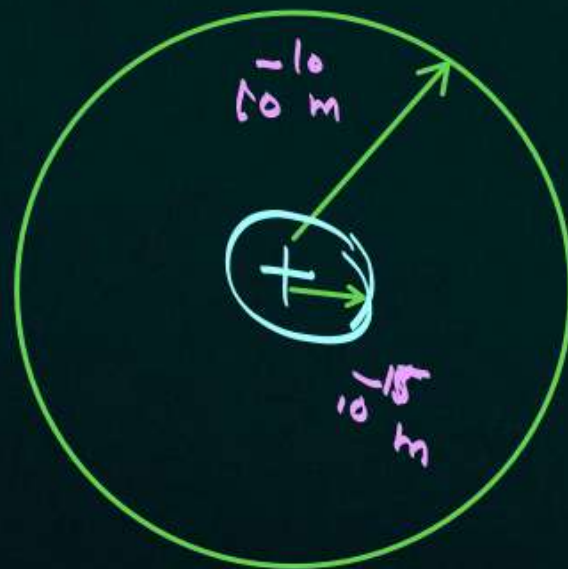


RUTHERFORD MODEL OF ATOM



C) CONCLUSION $\Rightarrow$

- i) Most of the space in an atom is empty, that's why almost all particles were passed through.
- ii) There is heavy positively charged particles present at the centre of atom, called nucleus that's why few α -particles were deviated & were sent back.
- iii) Radius of atom is 10^{-10} m
& Radius of nucleus 10^{-15} m.



POSTULATES $\Rightarrow$

- 1) Most of the space in an atom is empty and nucleus is present at the centre of atom.
- 2) Nucleus is at the centre & e^- revolve around the nucleus in a circular path called orbit.
- 3) There is an electrostatic force of attraction present between nucleus & electrons.

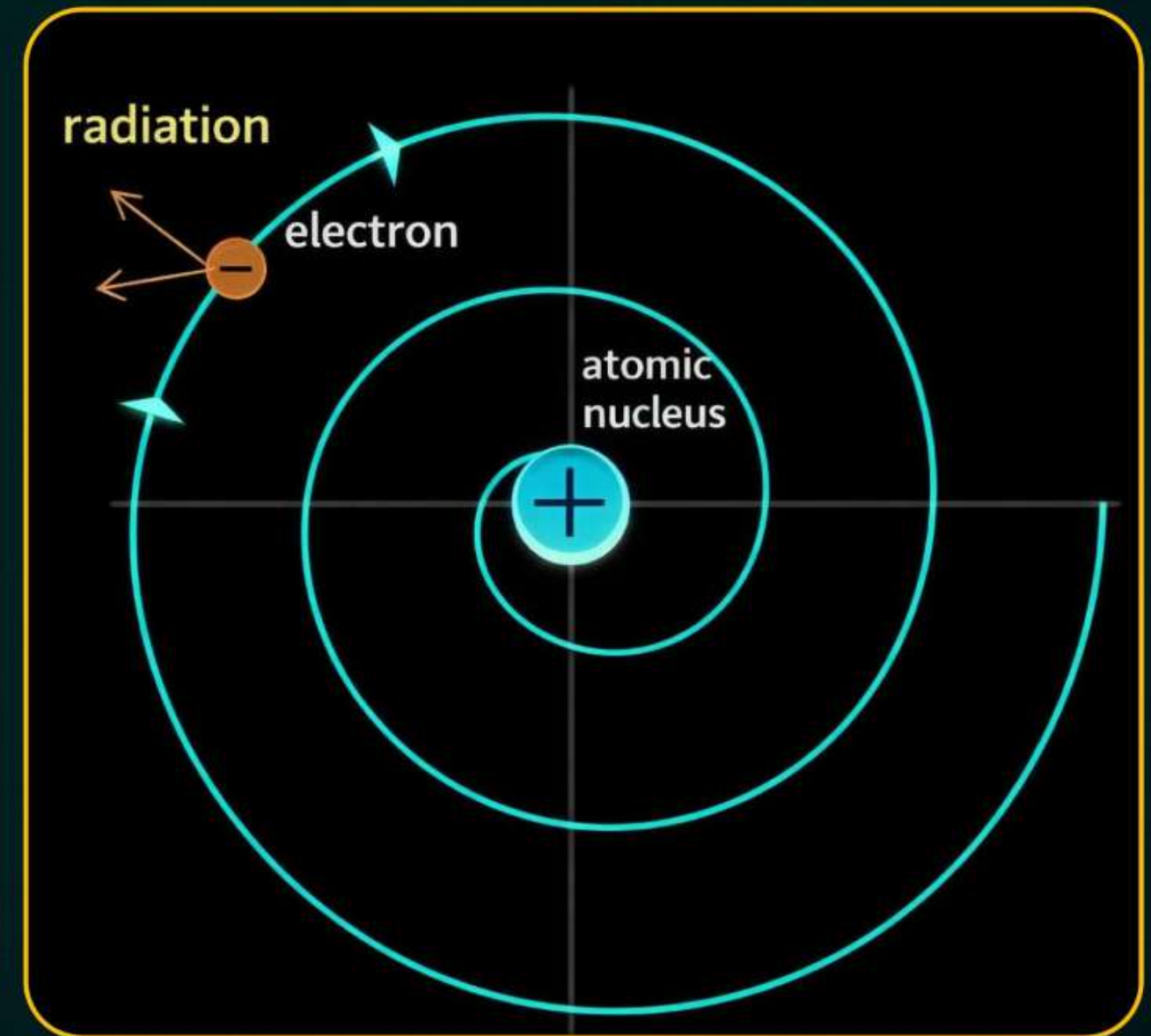


RUTHERFORD MODEL OF ATOM



LIMITATIONS:

- According to Maxwell's theory, a moving charged particle like an electron should give off energy all the time. As it loses energy, it should spiral into the nucleus and the atom should break down in just 10^{-8} seconds. But this never happens, showing that the theory doesn't fully explain how atoms work.
- This model does not explain structure of atoms i.e., the distribution of electrons around the nucleus and their energies.



ATOMIC NUMBER AND MASS NUMBER.

A) ATOMIC NUMBER $\rightarrow$ In a neutral atom, no. of e^- are always equal to no. of proton.

Atomic is defined as no. of protons present in any atom or ion. It is also called as proton number. Denoted by 'Z'.

B) Mass Number $\rightarrow$ It is sum of proton + neutron present in any atom or ion.

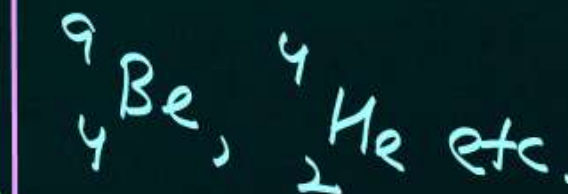
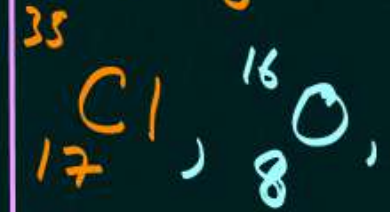
Proton + neutron is present in nucleus & why they are collectively called Nucleons. Mass no. is denoted by 'A'. mass of electron is neglected because it is very small.

i.e. $Z = \text{Atomic Number}$

$A = \text{Mass number}$

$$A = P + N \text{ or } Z + N.$$

Representation of an atom.



A

Z

Symbol.

QUESTION



Calculate the number of protons, neutrons and electrons in ${}_{35}^{80}\text{Br}$

Sol: $\rightarrow$ No. of Proton = 35

No. of electron = 35

we know that

$$A = P + N$$

$$80 = 35 + N.$$

$$80 - 35 = N.$$

QUESTION

NCERT

Homework for community



The number of electrons, protons and neutrons in a species are equal to 18, 16 and 16 respectively. Assign the proper symbol to the species.

Sol. Symbol

$$e = 18$$

$$p = 16$$

$$n = 16$$

QUESTION



What is the approximate radius of the nucleus according to Rutherford?

- 1 $10^{-18} m$
- 2 $10^{-10} m$
- 3 $10^{-15} m$
- 4 $10^{-5} m$

QUESTION

What is the mass number (A) of an atom with 6 protons and 7 neutrons?

- 1 6
- 2 7
- 3 12
- 4 13

$$A = P + N$$

Lallen

QUESTION



Which subatomic particles are present in the nucleus of an atom?

- 1 Electrons and Protons
- 2 Protons and Neutrons
- 3 Neutrons and Electrons
- 4 Electrons, Protons, and Neutrons



QUESTION



If an element has an atomic number of 15 and a mass number of 31, how many neutrons does it have?

1 15

2 16

3 31

4 46

QUESTION



H.W

In Rutherford's alpha particle scattering experiment, why do most alpha particles pass through the foil without deflection?

- 1 Atoms have a dense nucleus.
- 2 Atoms are mostly empty space.
- 3 Electrons occupy most of the space.
- 4 Alpha particles are positively charged.

QUESTION



Rutherford's experiment which established the nuclear model of the atom, used a beam of

- 1 β -particles, which impinged on a metal foil and got absorbed
- 2 γ -rays, which impinged on a metal foil and ejected electrons.
- 3 Helium atoms, which impinged on a metal foil and got scattered.
- 4 Helium nuclei, which impinged on a metal foil and got scattered

Thank
You